⚡ AutoRepoSync

> Never forget to pull again.

AutoRepoSync automatically keeps your local Git repositories synchronized with GitHub — silently in the background.

No prompts.
No manual pulls.
No "your branch is behind".

Just open VS Code and your repo stays updated.

🎬 Demo

AutoRepoSync Demo

Push → Fetch → Pull → Synced automatically.

🤯 The Problem

If you work in a team, you know this pain:

❌ You start coding
❌ Someone pushes new commits
❌ Your branch is now outdated
❌ You forget to pull
❌ Merge conflicts appear

Or worse:

Your branch is behind 'origin/main' by 5 commits

Now your workflow is broken.

💡 The Solution

AutoRepoSync handles everything automatically.

The extension runs in the background and:

1️⃣ Fetches remote changes
2️⃣ Detects if your branch is behind
3️⃣ Pulls updates automatically

4 Safely stashes your changes
5 Restores your work after syncing

You stay focused on coding.

✨ Features

Feature	Description
🔄 Auto Fetch	Periodically checks GitHub for updates
⚡ Auto Pull	Pulls updates automatically
📦 Auto Stash	Safely stashes local changes
🖥️ Dashboard	Real-time repo sync dashboard
📡 Webhook Mode	Instant sync from GitHub pushes
🗂️ Multi Repo	Works with multiple repos
🔔 Notifications	Optional commit notifications
📊 Activity Logs	Full sync history

🖥️ Dashboard

Monitor all repositories in one place.

Dashboard

Shows:

- Repo status
- Behind/ahead count
- Last sync
- Conflict alerts
- Activity logs

⚙️ How It Works

```
┌─────────────────┐
│ Timer/Event │
└─────────┬───────┘
          │
          ▼
┌─────────────────┐
│ git fetch │
└─────────┬───────┘
          │
```

```
▼
┌──────────────────┐
│ Compare │
│ Local/Remote│
└──────────┬───────┘
│
▼
┌──────────────────┐
│ git pull │
│ --rebase │
│ --autostash │
└──────────────────┘
```

🚀 Installation

Install from VS Code Marketplace

1. Open **VS Code**
2. Go to Extensions (`Ctrl + Shift + X`)
3. Search **AutoRepoSync**
4. Click **Install**

Install via VSIX

```
code --install-extension autoreposync.vsix
```

⚙️ Configuration

Customize the extension in VS Code settings.

Setting	Default	Description
autosync.enabled	true	Enable auto syncing
autosync.interval	30	Sync interval (seconds)
autosync.autoStash	true	Stash local changes
autosync.enableWebhookMode	false	Enable webhook sync
autosync.webhookPort	9090	Webhook server port
autosync.showNotifications	true	Show notifications

🎮 Commands

Command	Description

Command	Description
AutoRepoSync: Start Sync	Start auto sync
AutoRepoSync: Stop Sync	Stop auto sync
AutoRepoSync: Sync Now	Manual sync
AutoRepoSync: Open Dashboard	Open dashboard

📡 Webhook Mode

Enable instant syncing using GitHub Webhooks.

Steps:

1 Enable webhook mode in settings
2 Add webhook in GitHub repo
3 Auto sync triggers instantly on push

Payload URL:

http://localhost:9090/webhook

🏗 Architecture

```
src
├── extension.ts
├── gitManager.ts
├── syncService.ts
├── repoWatcher.ts
├── webhookServer.ts
└── dashboard
```

Modules:

- **Git Manager** → CLI wrapper
- **Sync Service** → orchestrates sync
- **Repo Watcher** → detects repositories
- **Dashboard** → monitoring UI
- **Webhook Server** → push-triggered sync

🛠 Development

Clone the repo:

```
git clone https://github.com/autoreposync/autoreposync.git
```

```
cd autoreposync
npm install
npm run compile
```

Run extension:

Press F5

📦 Package Extension

```
npm install -g @vscode/vsce
vsce package
```

Output:

autoreposync-1.0.0.vsix

📈 Roadmap

Planned features:

- 🤖 AI merge conflict resolution
- 📢 Slack / Discord alerts
- 📊 Team sync analytics
- 🧠 Smart branch awareness
- 📝 Commit summaries

🤝 Contributing

Pull requests are welcome.

Steps:

1. Fork repository
2. Create feature branch
3. Submit PR

⭐ Support

If this project helps you, please consider giving it a star ⭐

It helps more developers discover the project.

📄 License

MIT License